

Operating Systems: Assignment-1

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Q. 1 Practice basic linux commands- ls, cd, pwd, mkdir, rm, rmdir, touch, cp, mv, cat, chmod, kill, top, ssh, etc.

Command: ls

Output:

```
Desktop Documents Downloads Music Pictures Public Templates Videos snap
```

Command: cd Desktop

Output:

```
ubuntu@ubuntu:~/Desktop$
```

Command: pwd

Output:

```
/home/ubuntu
```

Command: mkdir test

Output:

```
ubuntu@ubuntu:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos snap test
```

Command: touch file.txt

Output:

```
ubuntu@ubuntu:~/test$ ls
file.txt
```

Command: cp file.txt file_new.txt

Output:

```
ubuntu@ubuntu:~/test1$ ls
file.txt file_new.txt
```

Command: mv file.txt file_new.txt

Output:

```
ubuntu@ubuntu:~/test1$ ls
file_new.txt
```

Command: rm file_new.txt

Output:

```
ubuntu@ubuntu:~/test$ ls
ubuntu@ubuntu:~/test$
```

Command: rmdir test

Output:

```
ubuntu@ubuntu:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  snap
```

Command: cat file_new.txt

Output:

```
Hello Linux
```

Command: chmod +x file_new.txt

Output:

```
total 4
-rwxrwxr-x 1 ubuntu ubuntu 12 Jan 10 20:03 file_new.txt
```

Command:

ps

kill 1234

Command: top

Output:

```
top - 06:14:05 up 4 min, 1 user, load average: 2.05, 2.57, 1.20
Tasks: 230 total, 2 running, 228 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.5 us, 0.5 sy, 0.0 ni, 92.7 id, 0.0 wa, 0.0 hi, 6.3 si, 0.0 st
MiB Mem : 3916.6 total, 108.3 free, 1297.8 used, 2816.6 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 2618.8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
2499	ubuntu	20	0	3903620	389380	149100	S	1.3	9.7	0:18.65	gnome-shell
2273	ubuntu	20	0	316540	97044	62536	S	0.7	2.4	0:08.87	Xorg
4639	ubuntu	20	0	23616	5984	3808	R	0.3	0.1	0:00.09	top
1	root	20	0	23504	14576	9712	S	0.0	0.4	0:04.43	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.02	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
9	root	20	0	0	0	0	R	0.0	0.0	0:13.07	kworker/0:0-events
10	root	20	0	0	0	0	I	0.0	0.0	0:00.39	kworker/0:1-cgroup_destroy
11	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri
12	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/u8:0-ipv6_addrconf
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
16	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
17	root	20	0	0	0	0	S	0.0	0.0	0:01.02	ksoftirqd/0
18	root	20	0	0	0	0	I	0.0	0.0	0:02.34	rcu_preempt
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_worker/0
20	root	20	0	0	0	0	S	0.0	0.0	0:00.08	rcu_exp_gp_kthread_worker
21	root	rt	0	0	0	0	S	0.0	0.0	0:00.02	migration/0
22	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0

Command: ssh user@192.168.1.10

Output:

```
ssh: connect to host 192.168.1.10 port 22: Connection refused
```

Q. 2 Create a folder using mkdir with the name "SVNIT". Create subfolders inside the SVNIT folder as AI, CSE, MED,ECE

1. Create two text files in a AI folder with name file1.txt and file1.txt

Command:

```
touch file1.txt file2.txt
```

Output:

```
ubuntu@ubuntu:~/SVNIT/AI$ ls
file1.txt  file2.txt
```

2. Copy file1.txt from AI folder to ECE Folder

Command:

```
cp file1.txt ../ECE/
```

Output:

```
ubuntu@ubuntu:~/SVNIT/AI$ ls
file1.txt  file2.txt
ubuntu@ubuntu:~/SVNIT/AI$ cd ..
ubuntu@ubuntu:~/SVNIT$ cd ECE
ubuntu@ubuntu:~/SVNIT/ECE$ ls
file1.txt
```

3. Move file1.txt from AI folder to MED Folder

Command:

```
mv file1.txt ../ECE/
```

Output:

```
ubuntu@ubuntu:~/SVNIT/AI$ ls
file2.txt
ubuntu@ubuntu:~/SVNIT/AI$ cd ..
ubuntu@ubuntu:~/SVNIT$ cd MED
ubuntu@ubuntu:~/SVNIT/MED$ ls
file1.txt
```

4. Change the permissions of file1.txt from ECE folder such that owner can perform read, write and execute where others can perform ONLY read

Command:

```
chmod 744 ../ECE/file1.txt
```

Output:

```
ubuntu@ubuntu:~/SVNIT/MED$ ls -l ../ECE/file1.txt
-rwxr--r-- 1 ubuntu ubuntu 0 Jan 11 06:21 ../ECE/file1.txt
```

5. Delete folder CSE

Command:

`rmdir CSE`

Output:

```
ubuntu@ubuntu:~/SVNIT$ ls  
AI  ECE  MED
```

Q.2 Practice Linux commands related to process management

1. Start a browser and check the process ID of the browser using Top command

Command:

```
firefox &  
top
```

Output:

```
top - 06:27:28 up 17 min, 1 user, load average: 1.44, 0.59, 0.64  
Tasks: 229 total, 1 running, 228 sleeping, 0 stopped, 0 zombie  
%Cpu(s): 24.3 us, 27.7 sy, 0.0 ni, 38.9 id, 0.0 wa, 0.0 hi, 9.1 si, 0.0 st  
MiB Mem : 3916.6 total, 133.7 free, 2055.8 used, 2345.1 buff/cache  
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 1860.8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
5310	ubuntu	20	0	10.9g	428068	162024	S	98.0	10.7	0:14.93	firefox
5702	ubuntu	20	0	10.4g	154648	87832	S	10.9	3.9	0:05.47	Privileged Cont
2273	ubuntu	20	0	331532	111636	73544	S	3.0	2.8	0:25.84	Xorg
27	root	20	0	0	0	0	S	2.3	0.0	0:06.00	ksoftirqd/1
1262	root	20	0	0	0	0	I	1.3	0.0	0:03.92	kworker/u9:7-loop7
2499	ubuntu	20	0	3972932	393748	152684	S	1.3	9.8	0:42.92	gnome-shell
1329	systemd+	20	0	22364	14232	11288	S	0.7	0.4	0:00.41	systemd-resolve
3895	ubuntu	20	0	850924	54604	43260	S	0.7	1.4	0:06.29	gnome-terminal-
5725	ubuntu	20	0	2476228	123988	83476	S	0.7	3.1	0:00.86	Isolated Web Co
5730	ubuntu	20	0	2420556	69220	54692	S	0.7	1.7	0:00.13	Web Content
6080	ubuntu	20	0	2420556	70148	54980	S	0.7	1.7	0:00.12	Web Content
18	root	20	0	0	0	0	I	0.3	0.0	0:03.00	rcu_preempt
57	root	20	0	0	0	0	S	0.3	0.0	0:01.38	kswapd0
67	root	20	0	0	0	0	I	0.3	0.0	0:04.24	kworker/u10:4-events_freezable_pwr_efficient
5659	ubuntu	20	0	2453992	97724	66700	S	0.3	2.4	0:00.56	WebExtensions
6103	ubuntu	20	0	307172	35480	25944	S	0.3	0.9	0:00.03	Utility Process
6109	ubuntu	20	0	2416436	68468	53940	S	0.3	1.7	0:00.08	Web Content
6439	ubuntu	20	0	23616	5860	3684	R	0.3	0.1	0:00.03	top
1	root	20	0	23504	14576	9712	S	0.0	0.4	0:04.62	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.02	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slab_flushwq

2. Kill the browser process using “kill” command

Command:

```
kill 5310
```

Output:

```
ubuntu@ubuntu:~$ Exiting due to channel error.  
[GFX1-]: CompositorBridgeChild receives IPC close with reason=AbnormalShutdown  
Exiting due to channel error.  
Exiting due to channel error.  
Exiting due to channel error.  
Exiting due to channel error.  
Exiting due to channel error.
```

3. Change the priority of any user process by using “nice” command

Command:

```
nice -n 10 sleep 300 &  
sudo renice -5 -p 2273
```

Output:

```
2273 (process ID) old priority 0, new priority -5
```